CHAPTER 7

COMPOSITION AND STRUCTURE OF ATMOSPHERE

CHAPTER ESSENCE AT A GLANCE:
Examine atmospheric gaseous composition, role of water vapour and aerosols, thermal stratification into troposphere, stratosphere, mesosphere, thermosphere, and exosphere, and the ozonosphere.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Gaseous Balance and Atmospheric Components:** Dry air is dominated by Nitrogen (78.08 percent) and Oxygen (20.95 percent); variable greenhouse components include Argon (0.93 percent), Carbon Dioxide (0.04 percent), ozone, water vapour (0 to 4 percent), and hygroscopic dust.

• **Tropospheric Convection and Normal Lapse Rate:** Lowest layer extending to 8 km at the poles and 18 km at the equator; contains almost all water vapour and weather phenomena; temperature decreases with altitude at the Normal Lapse Rate of 6.5 degrees Celsius per 1000 m.

• **Stratospheric Ozonosphere Barrier:** Extends from tropopause to 50 km; marked by temperature inversion due to absorption of harmful solar ultraviolet radiation by the ozone layer (O3), devoid of clouds and vertical convection.

• **Upper Atmospheric Layers:** Mesosphere (50 to 80 km, coldest layer reaching minus 100 degrees Celsius); Thermosphere and Ionosphere (80 to 400 km, containing electrically charged ions reflecting terrestrial radio waves); and outer Exosphere.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\Gamma = -\frac{dT}{dz} \approx 6.5\,^{\circ}\text{C per 1000 m} \rightarrow 0.65\,^{\circ}\text{C per 100 m (Normal Lapse Rate)}$$

Environmental Vertical Temperature Gradient in Troposphere

$$\frac{dP}{dz} = -\rho g \rightarrow \text{Hydrostatic Pressure Decline with Altitude}$$

Hydrostatic Law Governing Atmospheric Mass Distribution

$$O_2 + h\nu \rightarrow 2O, \quad O + O_2 \rightarrow O_3, \quad O_3 + h\nu \rightarrow O_2 + O \quad \text{(Ozone Cycle)}$$

Chapman Photochemical Stratospheric Ozone Equilibrium

$$P(z) = P_0 \exp\left(-\frac{Mgz}{RT}\right) \rightarrow \text{Exponential Pressure Attenuation}$$

Barometric Altitude Hypsometric Pressure Function

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The troposphere is significantly thicker at the equator (18 km) than at the poles (8 km) because strong solar heating at the equator creates intense vertical convectional currents that push air upwards.

• **Alert:** Temperature decreases with height in the troposphere and mesosphere, but increases with height in the stratosphere (due to ozone UV absorption) and thermosphere (due to shortwave solar radiation absorption).

• **Alert:** Dust particles and salt spray particles act as essential hygroscopic condensation nuclei around which atmospheric water vapour condenses to form cloud droplets and fog.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Name the two most abundant permanent gases of the atmosphere and state their percentage by volume. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Nitrogen (N2):** Accounts for approximately 78.08 percent of the total volume of dry air. **[1 Mark]**

• **Oxygen (O2):** Accounts for approximately 20.95 percent of the total volume of dry air. **[1 Mark]**

2. Why is the troposphere considered the most vital atmospheric layer for living organisms? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Weather Phenomena:** It contains virtually all atmospheric water vapour, clouds, and dust particles, hosting all life-sustaining weather phenomena like rainfall and storms. **[1 Mark]**

• **Breathable Air:** It contains over 75 percent of the total atmospheric mass and the dense oxygen necessary for terrestrial respiration. **[1 Mark]**

3. Explain the importance of the Ozone layer found in the Stratosphere. What causes its depletion? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Protective Shield:** The ozone layer absorbs lethal solar ultraviolet radiation, shielding living organisms from skin cancer, genetic mutations, and cataract damage. **[1.5 Marks]**

• **Ozone Depletion:** Caused by synthetic halocarbons, particularly Chlorofluorocarbons (CFCs) used in refrigerants and aerosols, which release free chlorine radicals that catalytically destroy ozone molecules. **[1.5 Marks]**

4. What is the Normal Lapse Rate? Calculate the temperature at an altitude of 2000 metres if sea-level temperature is 25 degrees Celsius. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** The rate at which atmospheric temperature decreases with increasing altitude in the troposphere, averaging 6.5 degrees Celsius per 1000 metres (or 1 degree Celsius per 165 metres). **[1.5 Marks]**

• **Calculation:** Temperature drop for 2000 m = 2 × 6.5 degrees Celsius = 13 degrees Celsius. Temperature at 2000 m = 25 - 13 = 12 degrees Celsius. **[1.5 Marks]**

5. Describe the five concentric thermal layers of the atmosphere based on temperature variations from the Earth's surface upwards. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Troposphere:** Extends to 8 km at poles and 18 km at equator; temperature decreases with height at normal lapse rate; bounded by tropopause. **[1 Mark]**

• **2. Stratosphere:** Extends up to 50 km; contains the protective ozone layer; temperature rises with altitude; bounded by stratopause. **[1 Mark]**

• **3. Mesosphere:** Extends from 50 km to 80 km; coldest layer where temperature drops to minus 100 degrees Celsius; meteors burn up here; bounded by mesopause. **[1 Mark]**

• **4. Thermosphere and Exosphere:** Thermosphere (80 to 400 km) contains ionosphere reflecting radio waves; temperature rises steeply; merges into outer exosphere. **[1 Mark]**

6. Explain the significance of water vapour and dust particles in atmospheric dynamics and weather systems.
[Score: 4 Marks]

Answer & Key Valuation Steps:

• **Water Vapour Significance:** Acts as a potent greenhouse gas absorbing terrestrial longwave radiation and retaining planetary warmth; supplies latent heat of condensation that fuels violent cyclones and storms. **[2 Marks]**

• **Dust Particles Significance:** Function as hygroscopic nuclei around which water vapour condenses to form clouds, dew, and fog; scatter incoming solar radiation producing red and orange hues at sunrise and sunset. **[2 Marks]**